PHARMACY QUERY ASSISTANT: OPEN-WEIGHT VS CLOSED LLM EVALUATION

project report submitted in partial fulfilment of the requirements for the Course

GENERATIVE AI AND PLATFORM ENGINEERING

BACHELOR OF TECHNOLOGY

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by

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DEPARTMENT OF CSE

Name of Title: Pharmacy Query Assistant: Open-Weight vs Closed LLM Evaluation

1. Abstract:

Large Language Models (LLMs) are now used in healthcare to answer medical questions, but they can give answers that sound confident and are still wrong. In pharmacy, a wrong answer about a dose, a drug interaction, or a contraindication can directly harm a patient. Most existing studies test only closed models such as ChatGPT on small sets of questions, do not use trusted drug sources, and rarely report cost or response time. To solve this problem, this project builds a Pharmacy Query Assistant that answers drug-related questions using Retrieval-Augmented Generation (RAG). In RAG, the system first finds the relevant passages from authoritative drug references, such as drug databases and product labels, and then gives them to the LLM, so the answer is based on real sources and can show where it came from. The project compares open-weight LLMs, which can be downloaded and run locally, with closed LLMs such as Gemini, which can be used only through an API. Open-weight models are attractive because patient queries can stay private, the models can be adapted to the task, and the cost per query is low, while closed models usually give higher accuracy. Both types are tested on the same set of pharmacy questions covering dosing, drug interactions, contraindications, adverse effects, and patient counselling, and the answers are checked against the reference sources. The models are compared using accuracy, relevance, hallucination rate, safety, response time, and cost per query. When the system cannot find enough evidence, it does not guess and advises the user to consult a pharmacist. This project shows which type of LLM is better suited for a reliable and safe pharmacy assistant, and whether privacy-friendly open models can be used in place of closed ones. The assistant is meant to support pharmacists and not to replace their professional judgement.

2. INTRODUCTION:

Generative AI has changed the way people create and use text, images, and data in many fields such as healthcare, finance, and education. Large Language Models (LLMs) are one of the main results of this change. An LLM is a Generative AI model that is trained on a very large amount of text, so it can read a question and write an answer in natural language. In healthcare, such models have already shown good results on medical exams and in answering patient questions.

However, a fluent answer is not always a correct answer. LLMs can produce statements that sound confident but have no support, which is called hallucination. Pharmacy is a very sensitive area for this problem, because a wrong statement about a dose, a drug interaction, or a contraindication can directly cause harm to a patient. Earlier studies that tested chatbots on drug questions found that the results were mixed, and that the chatbots could not replace conventional drug-interaction tools.

Most of these studies used closed models and small, ad hoc question sets, and few of them reported cost or response time.

LLMs are of two main types. Closed models, such as Gemini, can be used only through an API, and the user cannot see how they are built. Open-weight models, such as Llama and Mistral, publish their weights, so they can be downloaded, checked, and run on a local machine using tools such as Ollama or Hugging Face. This makes them useful when patient data must stay private. Retrieval-Augmented Generation (RAG) is a method that helps reduce hallucination. It first searches trusted sources, such as drug databases and product labels, for the passages that match the question, and then asks the model to answer using only those passages.

This project follows a simple process: collecting drug reference data, preparing it for search, retrieving the right passages for each question, generating answers with different LLMs, and checking the answers. Open-weight and closed LLMs are tested under the same conditions using accuracy, relevance, hallucination rate, safety, response time, and cost per query.

The main goal of this project is to build a retrieval-grounded Pharmacy Query Assistant and to find out how well open-weight LLMs perform compared with closed LLMs on pharmacy questions. The project is presented as decision support for pharmacists and not as a replacement for professional judgement.

3. SOFTWARE REQUIREMENTS:

This project is built using Python 3.x. It can be run on a local machine or in Google Colaboratory (Google Colab), depending on the option chosen for running the open-weight models. Google Colab is an online tool that lets us write and run Python code in a web browser without installing any software.

The project uses the following Python libraries and tools:

- **Pandas** – to load and clean the drug label data and the question set.
- **Sentence-Transformers** – to convert drug text and questions into embeddings for searching.
- **ChromaDB** – a vector database used to store the embeddings and retrieve the most relevant passages.
- **Gemini API (Google)** – to access the closed LLM, Gemini, used for comparison.
- **Ollama (Option 1)** – to download and run small open-weight LLMs (about 3B to 8B parameters) such as Llama, Mistral, or Gemma locally on the machine.
- **Hugging Face Transformers (Option 2)** – an alternative way to run the same open-weight LLMs, with 4-bit quantisation to fit limited memory.
- **Matplotlib (optional)** – to plot the comparison results.

Official drug label data from openFDA and DailyMed is used as the knowledge source. The system runs on a standard laptop with 16 GB RAM and no GPU, since small quantised open-weight models are used. An internet connection is needed for the Gemini API and for downloading data and models.

Signature of the Guide **Course Instructor**